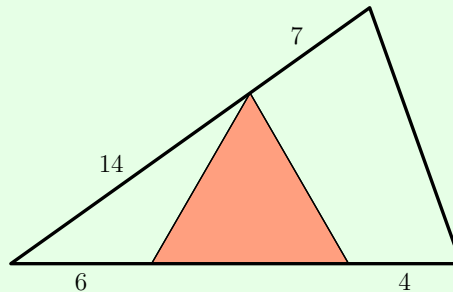


1 Easy

Problem 1 (Source: ???)

An equilateral triangle is placed inside a larger triangle splitting one side into lengths of 14 and 7 and another side into lengths of 6 and 4, as shown below.

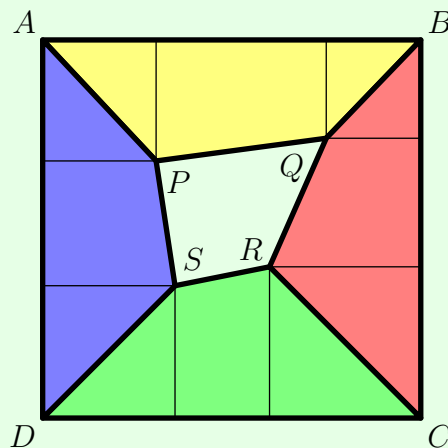


Find the area of the larger triangle.

2 Medium

Problem 2 (Source: ???)

Four squares are placed in each corner of a larger square, as shown in the diagram below. If the areas of quadrilaterals $APQB$, $BQRC$, $CRSD$ and $DSPA$ are 20, 23, 26 and 25, respectively, what is the area of quadrilateral $PQRS$?



3 Medium

Problem 3 (Source: ???)

In concave equilateral hexagon $HAUNTS$, we have $\angle A = 108^\circ$ and $\angle H = \angle U = 96^\circ$. Given that $\angle T < 180^\circ$, then compute $\angle T$.

4 Hard

Problem 4 (Source: ???)

Consider acute triangle $\triangle ABC$ with orthocenter H and circumcenter O such that $\triangle BCH$, $\triangle BCO$, and $\triangle ACO$ have areas 10, 13, and 14, respectively. Compute the area of $\triangle ABH$.

5 Hard

Problem 5 (Source: ???)

Consider triangle $\triangle ABC$ with circumcenter O and orthocenter H . The A -altitude meets BC at D and B -altitude meets AC at point E . Given that DE bisects OH , $AB = 20$, and $AC = 24$, find the sum of all possible values of BC .

6 Hard

Problem 6 (Source: ???)

Triangle $\triangle ABC$, with incenter I , has side lengths $AB = 30$, $BC = 70$, and $CA = 68$. The incircle touches BC , CA , and AB at D , E , and F , respectively. Let ℓ_1 be the line through D perpendicular to EF , and let ℓ_2 be the line through A perpendicular to BC . If $\ell_1 \cap \ell_2 = P$ and $PI \cap BC = Q$, find IQ^2 .